

Harisankar G

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Summary

Machine learning engineer experienced in building computer vision and deep learning systems for real-world datasets. Developed medical imaging segmentation models, activity detection pipelines, and generative design systems using PyTorch and TensorFlow. Skilled in data processing, model evaluation, and scalable ML experimentation using Python, Pandas, and scikit-learn.

Achievements

- IEEE Publication: *Machine Learning-Based Threat Detection in Crowded Environments*
- Amazon ML Challenge 2025: Ranked **192nd globally**
- Edge AI Hackathon 2025: Received Special Mention
- Solved 400+ algorithmic problems on LeetCode

Technical Skills

- **Programming:** Python, Java, R
- **Machine Learning:** CNNs, LSTM, Supervised Learning, Model Optimization, Hyperparameter Tuning
- **Libraries:** PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, OpenCV
- **Computer Vision:** Image Segmentation, Object Detection, Feature Extraction
- **Cloud & Tools:** AWS (EC2, S3 – basic deployment), Docker, Git
- **Systems:** Ubuntu, Windows, GPU-based training environments
- **Foundations:** Data Structures, Algorithms

Professional Experience

Data Analyst Intern — Zidio

Aug 2025 – Oct 2025

- Built LSTM-based forecasting models for financial time-series datasets.
- Developed preprocessing pipelines using Pandas and NumPy to clean and structure raw datasets.
- Trained and evaluated models using RMSE and MAE metrics.
- Automated data preparation workflows to accelerate experimentation.

Web Development Intern — TISE

Oct 2024 – Dec 2024

- Developed backend modules for structured data storage and retrieval.
- Implemented validation logic and data consistency checks.
- Contributed to debugging and feature updates in production code.

Projects

Brain Tumor Segmentation and Visualization System — Python, PyTorch, U-Net

- Built deep learning model for automatic segmentation of brain tumors from MRI scans.
- Implemented preprocessing, training, and evaluation pipelines.
- Achieved Dice score of **0.9284**.
- Developed visualization tools for easier interpretation of segmentation outputs.

Machine Learning-Based Threat Detection in Crowded Environments — Python, Computer Vision

- Designed computer vision pipeline combining YOLO and MediaPipe to detect suspicious activities.
- Processed video streams and extracted behavioral features.
- Achieved **99.1% AUC** and **92% F1-score**; published as IEEE research paper.

Generative AI for Automated Home Layout & Floor Planning — Python, Deep Learning

- Developed generative ML system producing building layouts from spatial constraints.
- Implemented training and evaluation pipelines.
- Achieved **98.2% constraint compliance**.

Education

Vellore Institute of Technology

Integrated MTech in Artificial Intelligence

Oct 2022 – Present

GPA: 8.17/10

Certifications

- **Applied Machine Learning in Python** — University of Michigan (Coursera) 2023
- **Cloud Computing** — NPTEL (IIT Kharagpur) 2024
- **Google Cloud Generative AI Virtual Internship** — Google Cloud 2025